



Universidad del Istmo

Facultad de Ingeniería

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Arquitectura de computadoras y microcontroladores

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Arquitectura de computadoras y microcontroladores

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Ingeniería en Sistemas y Ciencias de la Comunicación

Compilación :

Compilación con comando :

```
vcs -sverilog -full64 -R Top_view.sv mealy.sv moore.sv Caja_de_cambios_input.sv  
Clutch.sv decoder_7_segmentos.sv error_en_colocar_clutch.v next_state_logic.sv  
registro_estados.sv testbench1.sv
```

la compilación por medio de este comando es rápida pero a la hora de simularse tendrá que usar más tiempo para procesarse .

```
[vpezzarossi@synopsys-unis ~]$ vcs -sverilog -full64 -R Top_view.v moore.sv mealy.sv next_state_logic.sv registro_estados.sv error_en_colocar_clutch.v decoder_7_segmentos.sv Clutch.sv Caja_de_cambios_input.sv  
Chronologic VCS (TM)  
Version X-2025.06-SP2-6_Full64 -- Tue Sep 29 17:16:14 2026  
  
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Inclusivity & Diversity - Visit SolvNetPlus to read the "Synopsys Statement on Inclusivity and Diversity" (Refer to article 000036315 at https://solvnetplus.synopsys.com)  
  
Parsing design file 'Top_view.v'  
Parsing design file 'moore.sv'  
Parsing design file 'mealy.sv'  
Parsing design file 'next_state_logic.sv'  
Parsing design file 'registro_estados.sv'  
Parsing design file 'error_en_colocar_clutch.v'  
Parsing design file 'decoder_7_segmentos.sv'  
Parsing design file 'Clutch.sv'  
Parsing design file 'Caja_de_cambios_input.sv'  
Top Level Modules:  
  top_view  
TimeScale is 1 ns / 1 ps  
Starting vcs inline pass...  
2 modules and 0 UDP read.  
recompiling module top_view  
recompiling module decoder_7_segmentos  
Both modules done.  
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so
```

Compilación del Top como señal que abstrae los submodulos , KDB permite compilar el codigo entre veridi y synopsys :

```
vcs -full64 -sverilog -kdb -debug_access+all -top Testbench1 -o simv Top_view.sv  
mealy.sv moore.sv Caja_de_cambios_input.sv Clutch.sv decoder_7_segmentos.sv  
error_en_colocar_clutch.v next_state_logic.sv registro_estados.sv testbench1.sv
```

```
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Parsing design file 'decoder_7_segmentos.sv'  
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Top Level Modules:  
  top_view  
TimeScale is 1 ns / 1 ps  
Starting vcs inline pass...  
2 modules and 0 UDP read.  
recompiling module top_view  
recompiling module decoder_7_segmentos  
Both modules done.  
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so  
if [ -x ../simv ]; then chmod a-x ../simv; fi  
g++ -o ../simv -rdynamic -Wl,-rpath=$ORIGIN/ ../simv.daidir -Wl,-rpath=../simv.daidir -Wl,-rpath=/usr/synopsys/vcs/X-2025.06-SP2-6/linux64/lib -L/usr/synopsys/vcs/X-2025.06-SP2-6/linux64/lib -Wl,-rpath-link=/usr/lib64/libnuma.so.1 obj/simc.d.o _32837 archive1.so SIM.lo rmapats.mop.o rmapats.o rmar.o rmar.nd.o rmar.llvm 0 1.o rmar.llvm 0 0.o -lvrsim -lerrorinf -lsnpsmalloc -lvfs -lvcsnew -ldistsimclient -lsimprofile -luclinate /usr/synopsys/vcs/X-2025.06-SP2-6/linux64/lib/vcs_save_restore_new.o -ldl -lc -lm -lpthread -ldl  
../simv up to date  
Info: [VCS_SAVE_RESTORE_INFO] ASLR (Address Space Layout Randomization) is detected on the machine. To enable $save functionality, ASLR will be switched off and simv re-executed.  
Please use '-no_save' simv switch to avoid re-execution or '-suppress=ASLR_DETECTED_INFO' to suppress this message.  
Chronologic VCS simulator copyright 1991-2025  
Contains Synopsys proprietary information.  
Compiler version X-2025.06-SP2-6_Full64; Runtime version X-2025.06-SP2-6_Full64; Sep 29 17:16 2026  
V C S S i m u l a t i o n R e p o r t  
  
Time: 0 ps  
CPU Time: 0.280 seconds; Data structure size: 0.0Mb  
Tue Sep 29 17:16:16 2026  
CPU time: .365 seconds to compile + .338 seconds to elab + .204 seconds to link + .314 seconds in simulation  
[vpezzarossi@synopsys-unis ~]$
```

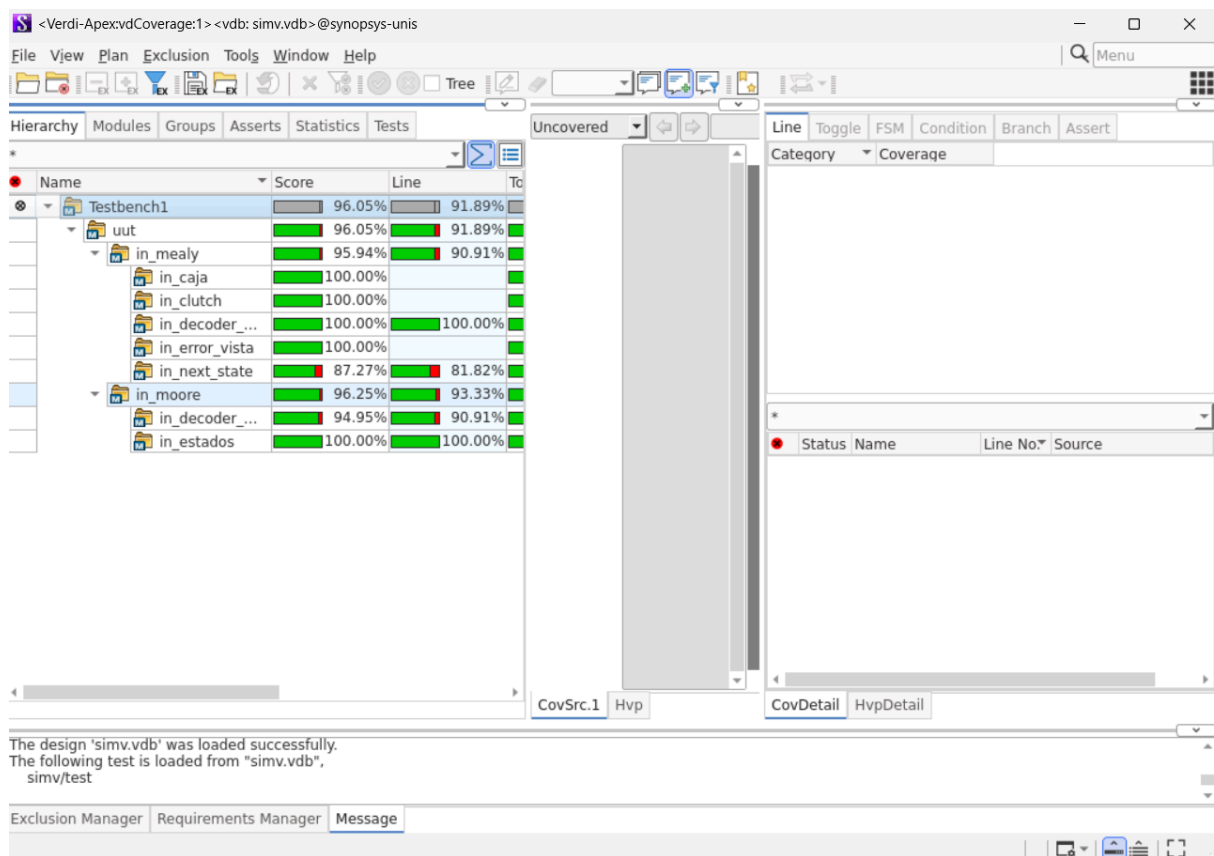
```
get -x ../simv ; then chmod a-x ../simv; fi  
git --no-simv -rdynamic -Wl,-rpath=:$ORIGIN/../lib -ldl -lc -lpthread -ldl  
sys/vcs/X-2025.06-SP2-6/linux64/lib -L/usr/synopsys/vcs/X-2025.06-SP2-6/linux64/lib -L/usr/syno  
pics/mop/rmapats.o rmar.nm.o rmar.lvm.0.o rmar.lvm.0.a  
../bin/mon/runsynopsys/vcs/X-2025.06-SP2-6/linux64/lib/ltoA.a -suclnactnew_tidsatnsynclicent -lsmpfr  
archive -lvscutil -Wl,-no-whole-archive -vcs.plt_stub.o /usr/synopsys/vcs/X-2025.06-SP2-6/linux64/lib/vcs_save_restore_new.o /usr/synopsys/verdi/Y-  
2026.03-SP2/share/PLI/VCS/LINUX64/pli.a -ldl -lc -lnl -lpthread -ldl  
../simv up to date  
CPU time : .338 seconds to compile + .315 seconds to elab + .334 seconds to link
```

```
vcs -full64 -sverilog -kdb -debug_access+all -top <tu archivo testbench> \
    -cm line+cond+branch+tgl+fsm \
    -cm_hier coverage.cfg \
    -o simv \
<tus archivos fsm>
```

Ejecutamos la simulación para que abstraiga la cobertura de cada test
./simv -no_save -cm line+cond+branch+tgl+fsm

Consultamos el reporte por medio de este comando
urg -dir simv.vdb -format both -report coverage_report

Visualización en GUI



Dashboard creado con los siguientes comandos :

Date: Thu Oct 1 00:46:42 2026
User: vpezzarossi
Version: X-2025.06-SP2-6
Command line: urg -dir simv.vdb -format both -report coverage_report
Number of tests: 1

Total Coverage Summary

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
96.05	91.89	100.00	100.00		92.31

Hierarchical coverage data for top-level instances

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	NAME
96.05	91.89	100.00	100.00		92.31	Testbench1

Total Module Definition Coverage Summary

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
96.61	92.31	100.00	100.00		94.12

Module Definition

Module self-instances :

NAME	SCORE	LINE	COND	TOGGLE	FSM	BRANCH
Testbench1.uut.in_mealy.in_clutch	100.00			100.00		

Module Instance :
Testbench1.uut.in_mealy.in_clutch

Instance :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
100.00			100.00		

Instance's subtree :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
100.00			100.00		

Parent :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	NAME
100.00			100.00			in_mealy

Subtrees :

NAME	SCORE	LINE	COND	TOGGLE	FSM	BRANCH
no children						

Since this is the module's only instance, the coverage report is the same as for the module.

Toggle

Toggle Coverage for Module : Clutch

	Total	Covered	Percent
Totals	2	2	100.00
Total Bits	4	4	100.00
Total Bits 0->1	2	2	100.00
Total Bits 1->0	2	2	100.00
Ports	2	2	100.00
Port Bits	4	4	100.00
Port Bits 0->1	2	2	100.00
Port Bits 1->0	2	2	100.00

Port Details

Name	Toggle	Toggle 1->0	Toggle 0->1	Direction
clutch_action	Yes	Yes	Yes	INPUT
clutch_resultado	Yes	Yes	Yes	OUTPUT

Module Definition

Testbench1.uut.in_mealy

Module Instance : Testbench1.uut.in_mealy

Instance :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
100.00			100.00		

Instance's subtree :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH
95.94	90.91	100.00	100.00		92.86

Parent :

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	NAME
100.00			100.00			uut

Subtrees :

NAME	SCORE	LINE	COND	TOGGLE	FSM	BRANCH
in_caja	100.00			100.00		
in_clutch	100.00			100.00		
in_decoder_mealy	100.00	100.00	100.00	100.00		100.00
in_error_vista	100.00			100.00	100.00	
in_next_state	87.27	81.82		100.00		80.00

Since this is the module's only instance, the coverage report is the same as for the module.

Toggle

Total Bits 1->0	22	22	100.00
Ports	7	7	100.00
Port Bits	40	40	100.00
Port Bits 0->1	20	20	100.00
Port Bits 1->0	20	20	100.00
Signals	1	1	100.00
Signal Bits	4	4	100.00
Signal Bits 0->1	2	2	100.00
Signal Bits 1->0	2	2	100.00

Port Details

Name	Toggle	Toggle 1->0	Toggle 0->1	Direction
palanca_de_cambio[1:0]	Yes	Yes	Yes	INPUT
clutch_cambio	Yes	Yes	Yes	INPUT
estado_actual[3:0]	Yes	Yes	Yes	INPUT
estado_siguiente[3:0]	Yes	Yes	Yes	OUTPUT
segmentos_mealy[6:0]	Yes	Yes	Yes	OUTPUT
led_error	Yes	Yes	Yes	OUTPUT
clutch_activo	Yes	Yes	Yes	OUTPUT

Signal Details

Name	Toggle	Toggle 1->0	Toggle 0->1
palanca_cambios[1:0]	Yes	Yes	Yes

3. Verdi y Forma de Ondas

Declaramos la siguiente instancia en nuestro código de testbench la cual creará un archivo .fsdb que es utilizado para las sondas de los circuitos, con este archivo podemos generar otros reportes que se basan en nuestro testbench.

initial begin

```
$fsdbDumpfile("dump.fsdb");
```

```
$fsdbDumpvars(0, <nombre de su testbench>, "+all");
```

end

Una vez creado el archivo de señales del testbench compilamos como top el testbench para que abstraiga nuestros submodulos de las FSM :

Paso 2 :

Compilan el archivo con este comando :

```
vcs -full64 -sverilog -kdb -debug_access+all -top <su archivo testbench> -o simv \
<sus archivos fsm y testbench>
```

Paso 3 :

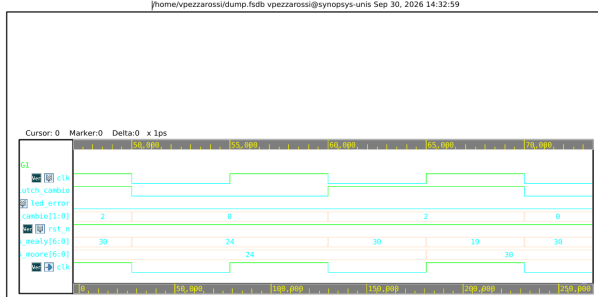
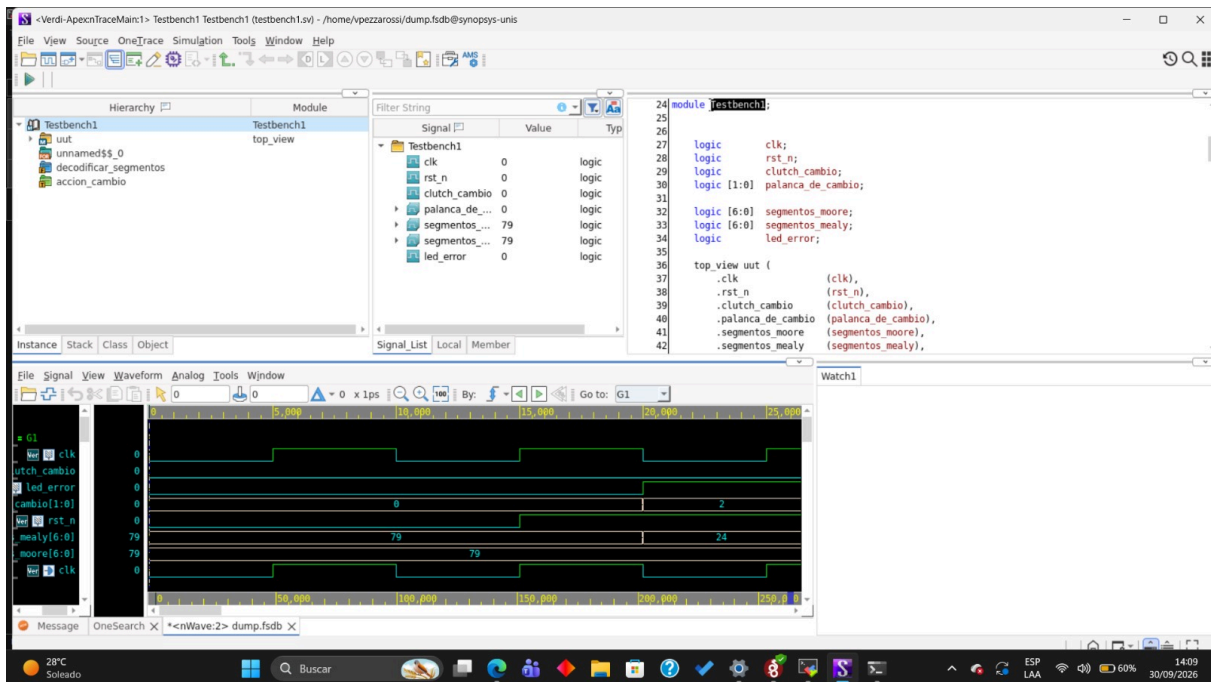
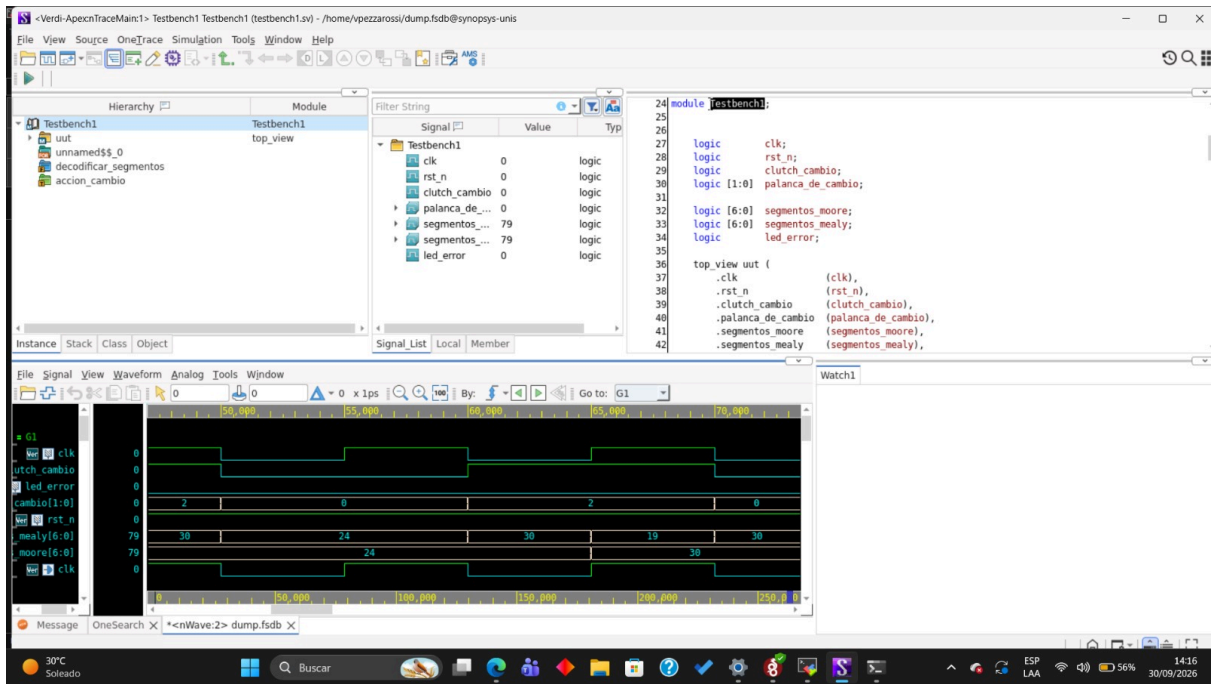
Corren la simulación

```
./simv -no_save
```

Paso 4 : cargan la simulación con las ondas

```
verdi -kdb -ssf dump.fsdb &
```

Schematic en GUI



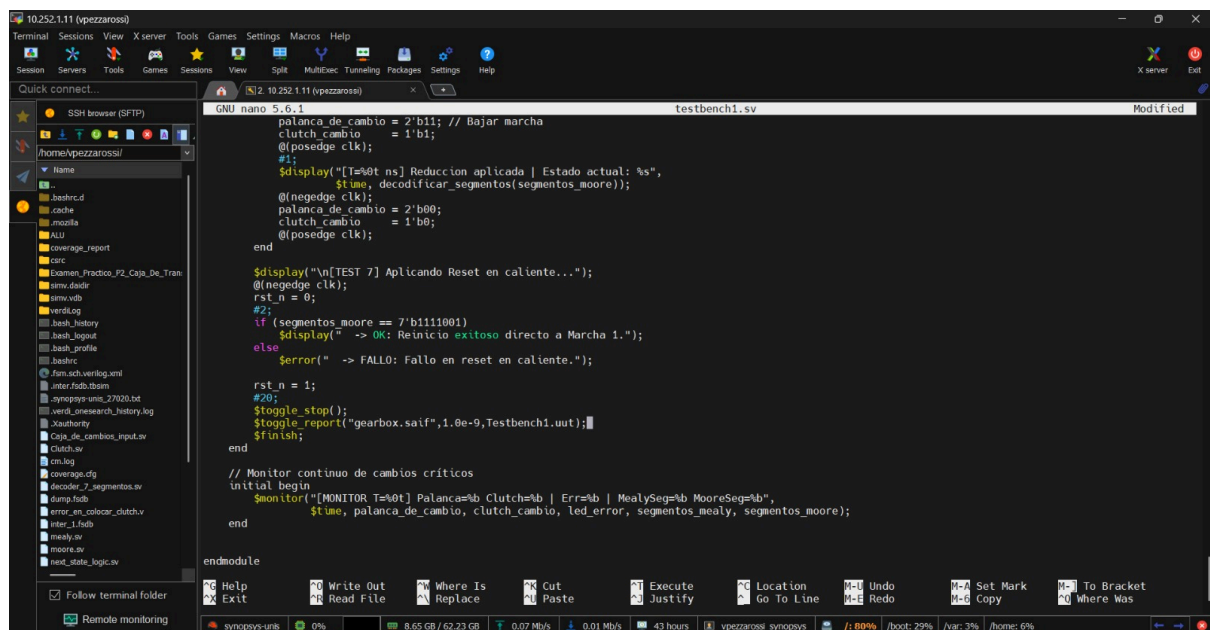
4. Archivo SAIF

Un archivo SAIF nos puede ayudar a recompilar la cantidad de veces que un circuito fue accionado , el tiempo que estuvo accionado en high y el tiempo el low para así ayudarnos a encontrar cuanto consume de recursos físicos y térmicos ese circuito dado a que esté operando de una forma nominal.

En nuestro Testbench podemos utilizar 2 instancias de toggle :

```
$set_toggle_region(top_tb.dut); $toggle_start();
```

nos ayudaran a verificar sobre cada modulo que esta ocurriendo , recopila la informacion y define la secuencia de pasos que ocurren en la simulación .



```
GNU nano 5.6.1 testbench1.v
// Bajar marcha
palanca_de_cambio = 2'b11;
clutch_cambio = 1'b1;
@(posedge clk);
#1;
$display("[T=%0t ns] Reduccion aplicada | Estado actual: %s",
$time, decodificar_segmentos(segmentos_moore));
@(negedge clk);
palanca_de_cambio = 2'b00;
clutch_cambio = 1'b0;
@(posedge clk);

end

$display("\n[TEST 7] Aplicando Reset en caliente...");
@(negedge clk);
rst_n = 0;
#2;
if (segmentos_moore == 7'b1111001)
$display(" -> OK: Reinicio exitoso directo a Marcha 1.");
else
$error(" -> FALLO: Fallo en reset en caliente.");

rst_n = 1;
#20;
$toggle_stop();
$toggle_report("gearbox.saif",1.0e-9,Testbench1.uut);
$finish;

end

// Monitor continuo de cambios criticos
initial begin
$monitor("[MONITOR T=%0t] Palanca=%b Clutch=%b | Err=%b | MealySeg=%b MooreSeg=%b",
$time, palanca_de_cambio, clutch_cambio, led_error, segmentos_mealy, segmentos_moore);
end

endmodule
```

```
$toggle_stop(); $toggle_report("cpu.saif", 1.0e-9, top_tb.dut);
```

Detiene la recopilación y guarda los datos en un archivo SAIF, siendo cada 1 pico segundo .

```
(DESIGN )
( DATE "Thu Oct 01 02:18:21 2026" )
( VENDOR "Synopsys, Inc." )
( PROGRAM_NAME "fsdb2saif" )
( VERSION "Verdi_Y-2026.03-SP2" )
( DIVIDER / )
( TIMESCALE 1 ps )
( DURATION 262000 )
( INSTANCE Testbench1
( NET
( clk
( T0 132000 ) ( T1 130000 ) ( TX 0 )
( TC 52 ) ( IG 0 )
)
( rst_n
( T0 17000 ) ( T1 245000 ) ( TX 0 )
( TC 3 ) ( IG 0 )
)
( clutch_cambio
( T0 162000 ) ( T1 100000 ) ( TX 0 )
( TC 20 ) ( IG 0 )
)
( palanca_de_cambio[1]
( T0 162000 ) ( T1 100000 ) ( TX 0 )
( TC 20 ) ( IG 0 )
)
)
)
)
```

TX : metaestado , si pasamos tiempo entre una señal 1 o 0